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Approximating discrete probability distributions with dependence trees

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Abstract

A method is presented to approximate optimally an n-dimensional discrete probability distribution by a product of second-order distributions, or the distribution of the first-order tree dependence. The problem is to find an optimum set of n - 1 first order dependence relationship among the n variables. It is shown that the procedure derived in this paper yields an approximation of a minimum difference in information. It is further shown that when this procedure is applied to empirical observations from an unknown distribution of tree dependence, the procedure is the maximum-likelihood estimate of the distribution.

Real-World Impact & Application

Smartphone Autocorrect & Predictive Text: Ever wonder how your phone keyboard predicts the next word you want to type? This paper introduced tree approximations of probabilities, allowing algorithms to predict likely combinations of words or features in milliseconds without draining your phone's battery.